

Cohit Barkun Bze

2018556053

Subject :

Date :/...../.....

1) General external merge sort ^{Horrible!!}
a) Number of passes: $1 + \lceil \log_{B-1} \lceil N/B \rceil \rceil$ ^{5-Pass Passy (11)}
Cost = $2N * (\# \text{ of passes})$

$$\text{Number of pages in file (N)} = \frac{150000}{(20/3)} = 22500$$

$$\text{1 page can store} = \frac{B}{R} = \frac{2000}{300} = \frac{20}{3}$$

$$\text{Number of passes} = 1 + \lceil \log_{10/3} \lceil 22500 / 11 \rceil \rceil = 5 //$$

$$\downarrow$$
$$\lceil 3.311 \rceil = 4$$

$$\text{Cost} = 2 * 22500 * 5 = 22500 //$$

$$\text{Time required} = S + r + b * b/t = 10 + 7 + (5 * 0.5) = 12 + 2.5 = 14.5 \text{ ms} //$$

b)

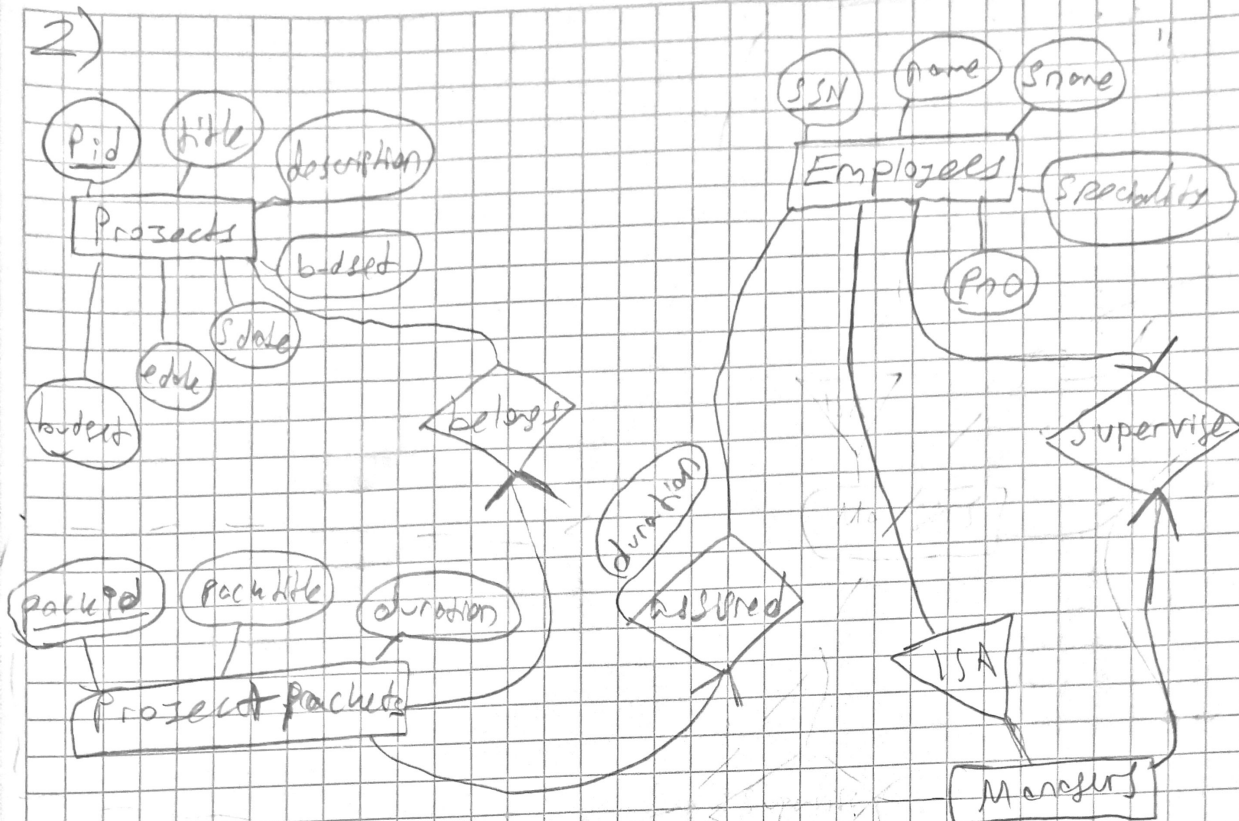
$$\text{Time} = 5 * (12.5) = 62.5$$

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2018556059

Date :/...../.....

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2)



3) PATIENTS(SSN, Name, Phone)

DRUGS (Barcode, Name, Description)

DOCTORS(SSN, Name, Specialty)

Prescription(SSN, Barcode, Amount, Date)

```

CREATE TABLE Patients (SSN CHAR(10),
                        name CHAR(30),
                        phone CHAR(10))
PRIMARY KEY(SSN)
    
```

```

CREATE TABLE Drugs (Barcode CHAR(10),
                     description CHAR(10),
                     name CHAR(30),
                     PRIMARY KEY(Barcode))
    
```

```

CREATE TABLE Doctors (ssn CHAR(10),
                       name CHAR(30),
                       specialty CHAR(10))
PRIMARY KEY(ssn)
    
```

```

CREATE TABLE Prescription(
    SSN CHAR(10),
    barcode CHAR(10),
    Amount INTEGER,
    Date Date,
    PRIMARY KEY(SSN, barcode)
    FOREIGN KEY(SSN)
    REFERENCES Patients
    ON DELETE CASCADE,
    FOREIGN KEY(barcode)
    REFERENCES Drugs)
    
```